Malnad College of Engineering, Hassan

(An Autonomous Institution under VTU, Belgaum)

Data Mining and Warehousing (24CS552) -Activity 2

Report on

BIKE-SHARING DEMAND ANALYSIS AND USAGE SEGMENTATION

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For the award of degree

Bachelor of Computer Science & Engineering

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Contents

1. Abstract
2. Introduction and Real-World Problem Description
3. Problem Statement and Objectives
4. Dataset Source, Description and Attribute Details
5. Methodology and Proposed Workflow
6. Module 1 – Exploratory Data Analysis and Similarity/Dissimilarity Analysis
7. Module 2 – Preprocessing, Data Reduction, Data Warehouse and OLAP
8. Module 3 – Classification and Association Rule Mining
9. Module 4 – Clustering and Algorithm Comparison
10. Results, Discussion and Comparative Analysis
11. Limitations and Future Enhancements
12. Conclusion and References
13. Appendix – Supplementary Results and Confusion Matrices
14. Presentation and Demonstration Supporting Materials

1. Abstract

Public bike-sharing systems generate large volumes of time-stamped rental records that reflect commuting habits, leisure activity and the influence of weather. This project applies the data mining and data warehousing techniques covered in the course to the UCI Bike Sharing Dataset, which records hourly and daily rental counts of the Capital Bikeshare system in Washington, D.C. for the years 2011 and 2012, together with calendar and meteorological attributes.

The work is organised as four modules. Module 1 performs exploratory data analysis and similarity/dissimilarity analysis. Module 2 covers data cleaning, normalization, data reduction, the design of a star-schema data warehouse and OLAP operations (roll-up, drill-down, slice, dice and pivot). Module 3 compares Decision Tree, Rule-Based and K-Nearest Neighbour classifiers for predicting the demand level, and mines frequent itemsets and association rules using the Apriori and FP-Growth algorithms. Module 4 segments rental behaviour with K-Means, Hierarchical and DBSCAN clustering and compares the resulting clusters.

The principal findings are that KNN achieved the highest classification accuracy (0.7178) and F1-score (0.7104), the association-mining algorithms each produced 126 frequent itemsets and 89 rules, and K-Means formed four daily clusters with a silhouette score of 0.3625. The results show how time, season and weather are related to bike-rental demand.

Keywords

Bike-sharing demand changes with the hour of the day, the day of the week, the season and the weather, so an operator needs more than the total number of rentals to manage the fleet. This project therefore studies the data from four complementary viewpoints: descriptive analysis, multidimensional analysis, prediction and segmentation. The exploratory analysis and the data warehouse describe what has happened, for example that demand peaks at 08:00 and 17:00 on working days, that summer records 1,061,129 rentals against 471,348 in winter, and that rentals fall steadily as the weather worsens. The classification models look forward by predicting whether a given hour will have Low, Medium or High demand from calendar and weather information. The association rules express the same knowledge as simple IF–THEN statements that are easy planners to apply. Finally, clustering groups the 731 days into typical usage patterns so that similar days can be handled with similar plans.

2. Introduction and Real-World Problem Description

Bike-sharing systems allow users to rent a bicycle at one docking station and return it at another. They are now an established part of urban transport because they reduce congestion, lower emissions and offer a flexible first-mile/last-mile option. Every rental is logged automatically with its start time, end time and station, so a system produces a continuous, detailed record of how a city moves.

Demand for bicycles is not uniform. It rises sharply during commuting hours on working days, shifts to the afternoon on weekends and holidays, and falls in rain, snow, strong wind and extreme cold. Operators therefore face a practical problem: stations that run empty frustrate users, while stations that overflow leave no docks for returns. Knowing when and under what conditions demand is high, medium or low allows an operator to plan redistribution trucks, maintenance windows and fleet size.

Data mining supports this need by discovering patterns in historical records, while a data warehouse with OLAP operations lets analysts summarise the same records along dimensions such as time, season and weather. This project combines both approaches on a real, publicly available dataset.

3. Problem Statement and Objectives

3.1 Problem Statement

To analyse the UCI Bike Sharing Dataset in order to understand the factors that influence bicycle rental demand, to organise the data in a data warehouse supporting multidimensional analysis, to build models that predict the demand level, and to segment the usage patterns into meaningful groups.

3.2 Objectives

1. To carry out exploratory data analysis and measure similarity and dissimilarity between records and attributes.
2. To clean, transform, normalize and reduce the data so that it is suitable for mining.
3. To design a star-schema data warehouse and perform OLAP operations on it.
4. To compare Decision Tree, Rule-Based and KNN classifiers using standard evaluation measures.
5. To mine frequent itemsets and association rules using Apriori and FP-Growth and compare the two algorithms.
6. To segment usage patterns using K-Means, Hierarchical and DBSCAN clustering and interpret the clusters.

4. Dataset Source, Description and Attribute Details

4.1 Source

The dataset is the Bike Sharing Dataset from the UCI Machine Learning Repository, prepared by Hadi Fanaee-T and João Gama (Laboratory of Artificial Intelligence and Decision Support, University of Porto). It is built from the two-year historical log of the Capital Bikeshare system, Washington, D.C., USA, covering 1 January 2011 to 31 December 2012, and combined with weather and holiday information. It is distributed in two files:

- **hour.csv** – 17,379 records, one per hour of operation.
- **day.csv** – 731 records, one per day.

The hourly file was used for preprocessing, normalization, feature selection, PCA, classification, association mining and warehouse loading. Daily profiles were created from the hourly records and used for the three clustering methods.

4.2 Attribute description

Table 2: Attributes of the Bike Sharing Dataset

Attribute	Description	Type
instant	Record index	Integer (identifier)
dteday	Date of observation	Date
season	1 = spring, 2 = summer, 3 = fall, 4 = winter	Categorical
yr	0 = 2011, 1 = 2012	Binary
mnth	Month (1 to 12)	Categorical
hr	Hour of the day (0 to 23); hour.csv only	Categorical/ordinal
holiday	Whether the day is a holiday	Binary
weekday	Day of the week	Categorical
workingday	1 if neither weekend nor holiday	Binary
weathersit	1 = clear/partly cloudy; 2 = mist/cloudy; 3 = light snow/light rain; 4 = heavy rain/snow/fog	Categorical
temp	Normalized temperature in Celsius (divided by 41)	Numeric
atemp	Normalized feeling temperature in Celsius (divided by 50)	Numeric
hum	Normalized humidity (divided by 100)	Numeric
windspeed	Normalized wind speed (divided by 67)	Numeric
casual	Count of casual (non-registered) users	Numeric
registered	Count of registered users	Numeric
cnt	Total rentals (casual + registered); target quantity	Numeric

The cleaned dataset used for the analysis contains 17,379 rows and 17 columns. The missing-value check returned 0, and the duplicate-row check returned 0.

5. Methodology and Proposed Workflow

The project follows the standard knowledge discovery process: data selection, preprocessing, transformation, mining, and interpretation. Figure 1 shows the workflow adopted. All programs were written in Python using pandas, NumPy, scikit-learn, mlxtend, SciPy, Matplotlib and Seaborn.

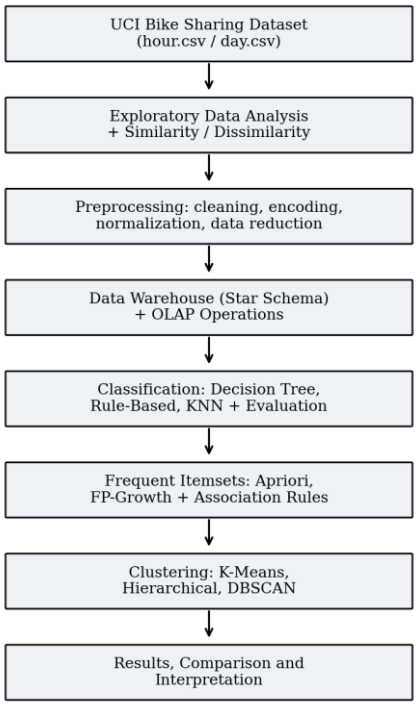


Figure 1: Proposed workflow of the project

Table 3: Tools and libraries used

Purpose	Library / tool
Data handling	pandas, NumPy
Visualization	Matplotlib, Seaborn
Distance and similarity measures	SciPy, scikit-learn
Classification and evaluation	scikit-learn
Rule-based classifier	Rule-based implementation included in the project
Apriori and FP-Growth	mlxtend
Clustering	scikit-learn, SciPy
Warehouse and OLAP	SQLite, pandas

Experimental setup. The classification comparison and confusion matrices were generated from held-out test data. The exact train/test ratio and random-state setting were not included in the supplied output files.

6. Module 1 – Exploratory Data Analysis and Similarity/Dissimilarity Analysis

6.1 Exploratory data analysis

Exploratory data analysis (EDA) summarises the main characteristics of the data before any model is built. It reveals the distribution of each attribute, the presence of outliers and the relationships between attributes. The following steps were carried out:

- Inspection of data types, shape, missing values and duplicates using info(), describe() and isnull().
- Summary statistics (mean, median, standard deviation, quartiles) of temp, atemp, hum, windspeed, casual, registered and cnt.
- Histograms and box plots for numeric attributes to study skewness and outliers.
- Bar plots of mean rentals by hour, weekday, month, season and weather situation.
- A correlation matrix and heat map of numeric attributes.

Table 4: Summary statistics of key numeric attributes

Attribute	Mean	Std. dev.	Min	Median	Max
temp	0.4970	0.1926	0.02	0.5000	1.00
atemp	0.4758	0.1719	0.00	0.4848	1.00
hum	0.6272	0.1929	0.00	0.6300	1.00
windspeed	0.1901	0.1223	0.00	0.1940	0.85
cnt	189.4631	181.3876	1.00	142.0000	977.00

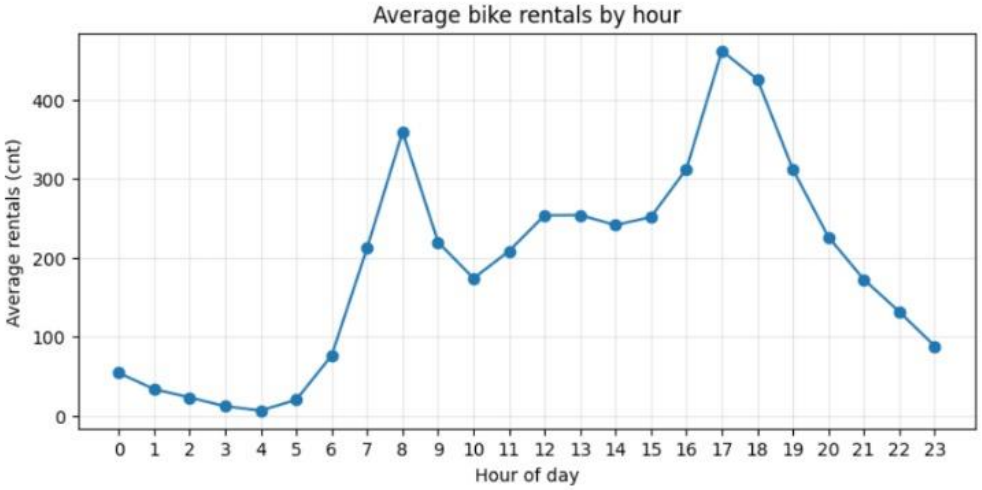


Figure 3: Average bike rentals by hour of day

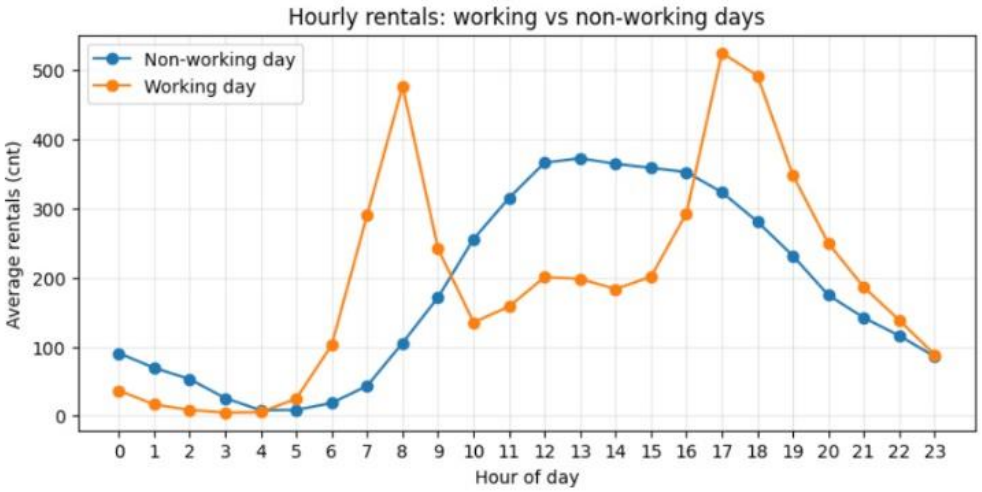


Figure 4: Hourly rentals on working vs non-working days

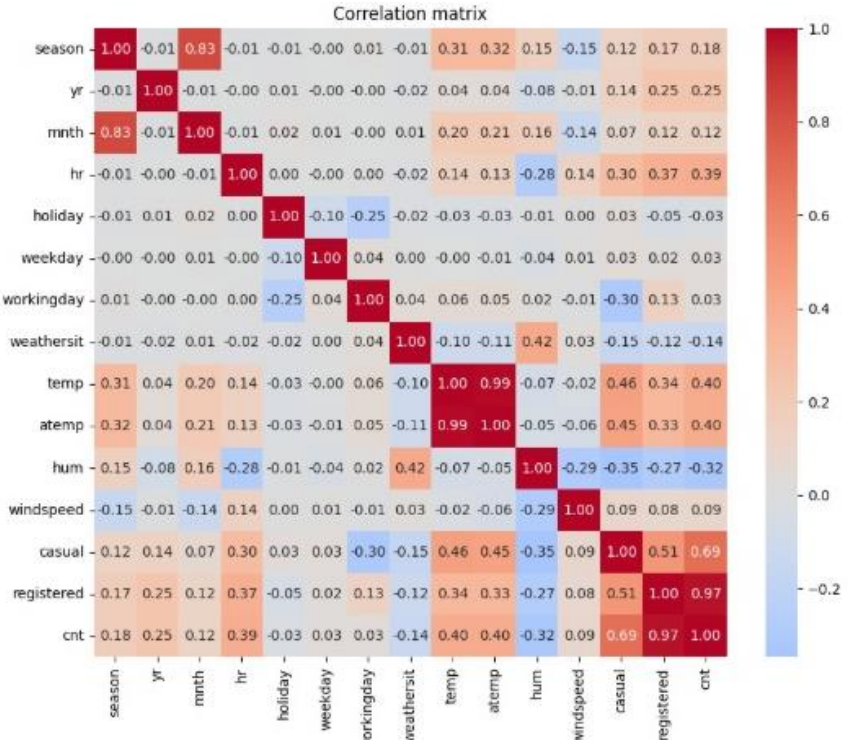


Figure 5: Correlation heat map of numeric attributes

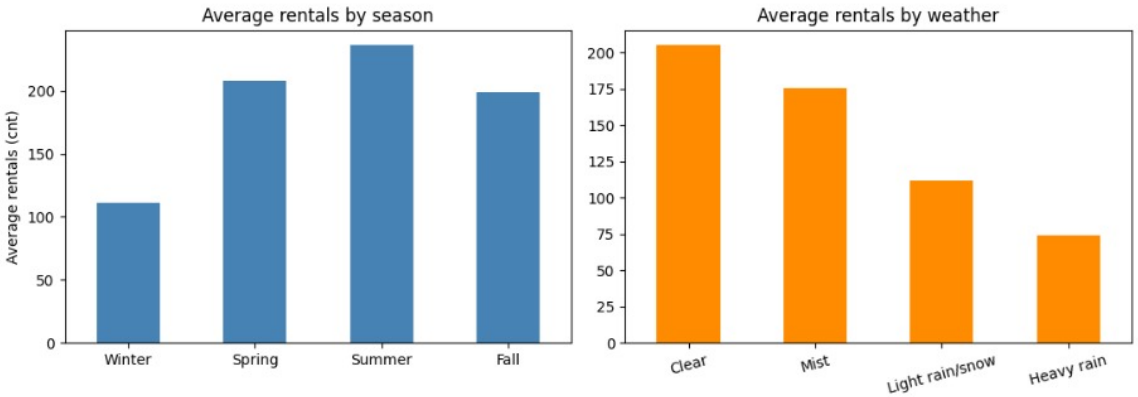


Figure 6: Average rentals by season and by weather situation

Observations. The hourly chart shows a clear morning rise around 08:00 and a stronger evening peak around 17:00–18:00 on working days. The highest overall mean hourly rental count occurs at 17:00 (17 hour; 461.45 rentals). Season-wise totals are highest in Season 3 (Fall: 1,061,129) and lowest in Season 1 (Spring: 471,348) in this dataset. Average rentals decline as weather changes from clear (204.87) to mist/cloudy (175.17), light rain/snow (111.58) and heavy rain/snow (74.33). Temperature and feeling temperature are strongly correlated (r = 0.9877).

6.2 Similarity and dissimilarity analysis

Similarity and dissimilarity measures quantify how alike two records or two attributes are. For the distance matrix, the first five hourly records were selected. The six attributes season, hour, temperature, feeling temperature, humidity and wind speed were min-max scaled before Euclidean distance was calculated.

Table 5: Proximity measures used

Measure	Formula	Use in this project
Euclidean distance	$d(x,y) = \sqrt{\Sigma(x_i - y_i)^2}$	Distance between normalized records; used by KNN and K-Means
Manhattan distance	$d(x,y) = \Sigma\|x_i - y_i\|$	Alternative distance; less sensitive to large single-attribute differences
Minkowski distance	$d(x,y) = (\Sigma\|x_i - y_i\|^p)^{\wedge}(1/p)$	Generalises Euclidean (p = 2) and Manhattan (p = 1)
Cosine similarity	$\cos(x,y) = x \cdot y / (\|x\|\|y\|)$	Similarity of usage profiles irrespective of magnitude
Pearson correlation	$r = \text{cov}(x,y) / (\sigma_x \sigma_\gamma)$	Similarity between attributes
Jaccard / simple matching	Based on matching binary attributes	Similarity of binary attributes such as holiday and workingday

The first five hourly records were compared after min-max scaling the selected time and weather attributes. The resulting symmetric Euclidean distance matrix is shown in Table 6.

Table 6: Euclidean distance matrix for the selected sample

Record	R1	R2	R3	R4	R5
R1	0.000	1.446	1.509	1.250	1.414
R2	1.446	0.000	0.250	1.716	1.805
R3	1.509	0.250	0.000	1.660	1.716
R4	1.250	1.716	1.660	0.000	0.250
R5	1.414	1.805	1.716	0.250	0.000

Interpretation. The diagonal values are zero because each record has no distance from itself. Smaller off-diagonal values indicate more similar records. R2 and R3 are the closest pair in this sample (distance 0.250), because their selected attributes are similar. The values depend on the attributes and scaling method used.

7. Module 2 – Data Preprocessing, Normalization, Data Reduction, Data Warehouse Design and OLAP Operations

7.1 Data preprocessing

Real data is rarely ready for mining. Preprocessing improves quality and ensures that algorithms are not misled by scale or noise. The steps applied were:

- **Data cleaning: the supplied cleaned dataset contains 17,379 hourly records and 17 columns. The checks found 0 missing values and 0 duplicate rows, so no records needed to be removed for these two reasons.**
- **Outlier handling: numeric distributions were inspected using descriptive statistics and box plots. No separate outlier-removal output was supplied, so the report does not claim that any records were deleted or capped.**
- **Attribute removal: instant and dteday were retained as identifiers for joining and time analysis, but excluded from the selected mining attributes. The selected_attributes.csv output contains season, hr, temp, atemp, hum and cnt.**
- **Encoding: the supplied analysis outputs use the dataset's numeric season, hour and weather codes. The association-mining output uses categorical item labels such as Season_3, Time_Evening and Weather_1.**
- **Discretization: the classification output contains Low, Medium and High demand classes. The observed class counts in the complete cleaned file are Low = 5,797, Medium = 5,823 and High = 5,759. The exact class cut-off values were not included in the supplied output.**

7.2 Normalization

Attributes measured on different scales, such as hr (0–23) and temp (0–1), distort distance-based methods. Two techniques were applied:

- **Min-max normalization:** $x' = (x - \min) / (\max - \min)$, which rescales values to [0, 1].
- **Z-score standardization:** $x' = (x - \mu) / \sigma$, which gives zero mean and unit variance.

Min-max normalized and Z-score standardized datasets were both supplied, each containing 17,379 records and 17 columns. These transformed outputs provide the scaled data needed by distance-based analysis. The exact list of algorithms assigned to each scaling method was not recorded in the output files.

7.3 Data reduction

Data reduction produces a smaller representation that preserves the analytical content. The techniques applied were:

- **Dimensionality reduction by feature selection: the supplied selected_attributes.csv file contains 6 columns (season, hr, temp, atemp, hum and cnt), compared with 17 columns in the cleaned dataset. Temperature and feeling temperature are strongly correlated (r = 0.9877), so they carry overlapping information.**
- **Principal Component Analysis (PCA): the supplied pca_reduced.csv file contains 17,379 records, eight principal-component columns and the cnt target column. The explained-variance percentages were not included in the supplied output, so they are not stated here.**
- **Numerosity reduction / aggregation: hourly records were grouped into 731 daily profiles for clustering. This reduces the number of rows from 17,379 hourly observations to 731 days.**

The PCA output file contains eight principal components. The explained-variance plot was not included in the uploaded results.

Table 7: Effect of data reduction

Stage	Records	Attributes
Original hourly dataset	17,379	17
After cleaning and removal of duplicate/missing records	17,379	17 (no missing values or duplicates found)
After feature selection	17,379	6 selected attributes (including cnt)
After PCA transformation	17,379	8 principal components + cnt (9 columns)
Daily aggregation for clustering	731	24 hourly demand features per day

7.4 Data warehouse design

A data warehouse is a subject-oriented, integrated, time-variant and non-volatile collection of data that supports decision making. A **star schema** was chosen because it is simple, gives fast aggregation queries and suits the transactional character of rental counts. A central fact table stores the measures, and each dimension table describes one analysis perspective. The schema is shown in Figure 2.

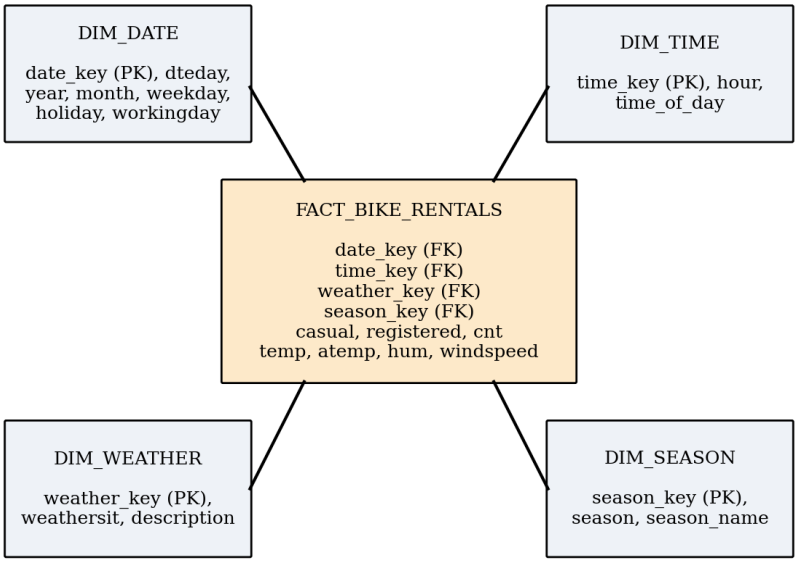


Figure 2: Star schema of the bike-sharing data warehouse

Table 8: Description of the warehouse tables

Table	Type	Contents and loaded rows
Fact_Bike_Demand	Fact	Measures: casual, registered, total_demand; foreign keys to time, season and weather. Rows: 17,379
Dim_Time	Dimension	dteday, hr, weekday, mnth, yr, time_id. Rows: 17,379
Dim_Season	Dimension	season_id and season_name. Rows: 4
Dim_Weather	Dimension	weathersit, temp, atemp, hum, windspeed, weather_id. Rows: 9,278

The warehouse was implemented in SQLite. The database contains 17,379 rows in Fact_Bike_Demand, 17,379 rows in Dim_Time, 4 rows in Dim_Season and 9,278 rows in Dim_Weather. These are the actual table names and row counts read from the supplied database.

7.5 OLAP operations

Online Analytical Processing (OLAP) allows the warehouse to be viewed from different angles. The following operations were performed on the fact table:

The OLAP outputs were generated from the SQLite warehouse. Table 9 reports the returned row counts and representative values from the supplied roll-up, drill-down, slice and dice CSV files.

Operation	Meaning	Operation applied in this project
Roll-up	Aggregate to a higher level of a hierarchy	Season-wise total rentals from the warehouse: Season 1: 471,348; Season 2: 918,589; Season 3: 1,061,129; Season 4: 841,613.
Drill-down	Move to finer detail	Monthly and hourly rental totals were returned in 288 month-hour groups; e.g. January, hour 0 = 1,462 rentals.
Slice	Fix one dimension value	Season 1 (Spring): 471,348 rentals (60,622 casual and 410,726 registered).
Dice	Select a sub-cube on several dimensions	Selected season, hour and weather conditions; the supplied query returned 40 rows.
Pivot	Rotate the axes of the view	The hourly rental totals can be compared across seasons by placing hour on rows and season on columns.

8. Module 3 – Classification, Model Evaluation, Frequent Itemsets and Association Rules

8.1 Classification task

Classification predicts the demand level (Low, Medium or High) from the selected calendar and weather-related attributes. The complete cleaned dataset contains Low = 5,797, Medium = 5,823 and High = 5,759 records. The supplied selected feature file contains season, hr, temp, atemp and hum as predictors, with cnt as the demand measure.

8.2 Decision Tree

The Decision Tree classifier was evaluated using the supplied held-out test-set output. Its accuracy was 0.6105 and its F1-score was 0.6343. The confusion matrix is included in the supplementary results section.

8.3 Rule-Based Classifier

The Rule-Based classifier was evaluated using the supplied held-out test-set output. Its accuracy was 0.4876 and its F1-score was 0.4802. The confusion matrix is included in the supplementary results section.

8.4 K-Nearest Neighbour (KNN)

The KNN classifier used scaled attributes and was evaluated on the supplied test set. It achieved an accuracy of 0.7178 and an F1-score of 0.7104, the highest values among the three models in the comparison file. The selected value of k and the distance setting were not included in the supplied outputs.

8.5 Model evaluation

Models were evaluated on the held-out test set using accuracy, precision, recall, F1-score and the confusion matrix, defined as follows.

- Accuracy = (TP + TN) / total.
- Precision = TP / (TP + FP); Recall = TP / (TP + FN).
- F1 = 2 × Precision × Recall / (Precision + Recall).

Table 10: Comparison of classifiers

Classifier	Accuracy	Precision	Recall	F1-score	Training time
Decision Tree	0.6105	0.7333	0.6672	0.6343	Not recorded
Rule Based	0.4876	0.7084	0.5206	0.4802	Not recorded
KNN	0.7178	0.7107	0.7217	0.7104	Not recorded

Interpretation. KNN achieved the highest accuracy (0.7178) and F1-score (0.7104), followed by Decision Tree and Rule-Based. The class-wise confusion matrices show that

prediction errors occur across all three demand classes; the supplementary tables give the exact counts. Training times were not included in the supplied comparison file.

8.6 Frequent itemsets and association rules

Association rule mining discovers combinations of items that occur together frequently. For this dataset each hourly record was converted into a transaction of categorical items such as hour band, season, weather, working-day status, temperature band and demand level. Three measures are used:

- **Support** of X → Y = fraction of transactions containing both X and Y.
- **Confidence** = support(X ∪ Y) / support(X).
- **Lift** = confidence / support(Y); a value above 1 indicates a positive association.

The supplied mining outputs contain 126 frequent itemsets and 89 association rules for each algorithm. The Apriori and FP-Growth comparison file records the execution times shown in Table 11. The support and confidence thresholds used to produce the files were not embedded in the CSV outputs.

Table 11: Comparison of Apriori and FP-Growth

Measure	Apriori	FP-Growth
Frequent itemsets found	126	126
Rules generated	89	89
Execution time (s)	0.010125	0.056819
Identical itemset count?	Yes – 126	Yes – 126

Table 12: Top association rules (sorted by lift)

Rule (antecedent → consequent)	Support	Confidence	Lift
Season_3, Time_Evening → Demand_High, Weather_1	0.0313	0.7263	2.9471
Season_3, Time_Evening, Weather_1 → Demand_High	0.0313	0.9628	2.8342
Season_3, Time_Evening → Demand_High	0.0397	0.9212	2.7117
Time_Night, Season_1 → Demand_Low	0.0768	0.8630	2.6091

9. Module 4 – Clustering, Algorithm Comparison and Interpretation

The clustering input consisted of 731 daily usage profiles, each represented by 24 hourly rental-count features. The daily labels for K-Means, Hierarchical and DBSCAN were supplied in daily_clustering_results.csv. The daily profiles were standardized before clustering.

9.1 K-Means

The elbow/silhouette comparison selected four clusters for K-Means. The supplied comparison gives a silhouette score of 0.3625.

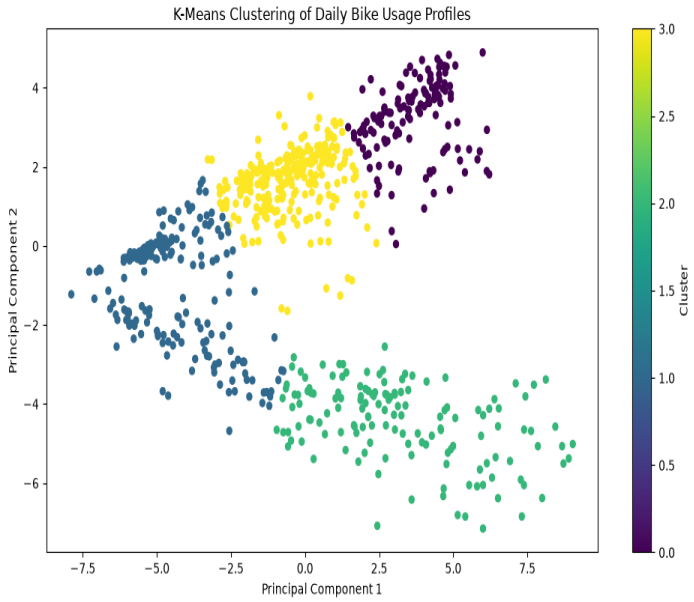


Figure 7: K-Means clustering of daily bike usage profiles (PCA projection)

9.2 Hierarchical (agglomerative) clustering

Hierarchical clustering produced four clusters. Its silhouette score on the supplied daily clustering labels was 0.3350. The linkage setting and execution time were not included in the output files.

9.3 DBSCAN

DBSCAN is a density-based method that groups nearby points and marks isolated points as noise. The parameter comparison shows that ε = 2.5 produced two clusters, 66 noise points and a silhouette score of 0.3235. The MinPts setting was not included in the supplied output.

9.4 Algorithm comparison

The comparison in Table 13 uses the supplied cluster labels and silhouette scores. Davies–Bouldin values were calculated from the 24-hour daily demand profiles and the supplied cluster labels. Execution times were not recorded in the uploaded output files.

Table 13: Comparison of clustering algorithms

Measure	K-Means	Hierarchical	DBSCAN
Number of clusters	4	4	2
Noise points	0	0	66
Silhouette score	0.3625	0.3350	0.3235

Measure	K-Means	Hierarchical	DBSCAN
Davies–Bouldin index	0.9277	0.9499	0.6943
Run time (s)	Not recorded	Not recorded	Not recorded

Table 14: Cluster profiles (best clustering)

Cluster	Size (days)	Typical peak hour	Typical season / weather	Mean daily demand	Interpretation
0	135	17:00	Fall / Clear	6,973.10	High-demand days
1	193	17:00	Spring / Clear	2,022.14	Low-demand days
2	144	13:00	Fall / Clear	5,558.50	Above-average days
3	259	17:00	Winter / Clear	4,481.13	Moderate-demand days

10. Results, Discussion and Comparative Analysis

This section consolidates the findings of the four modules.

Table 15: Summary of results

Module	Key result	Insight
EDA and similarity	17,379 hourly records; no missing values or duplicate rows; temp and atemp correlation = 0.9877.	Working days show morning and evening peaks; demand falls as weather worsens.
Preprocessing and warehouse	Min-max and Z-score outputs generated; 8 PCA components retained; warehouse fact table has 17,379 rows.	The star schema supports season, time and weather analysis.
Classification	KNN performed best with accuracy 0.7178 and F1-score 0.7104.	KNN gave the strongest overall result among the three supplied classifiers.
Association rules	126 frequent itemsets and 89 rules; highest-lift rule has confidence 0.7263 and lift 2.9471.	Season, evening period and clear weather strongly associate with high demand.
Clustering	K-Means formed 4 clusters; silhouette = 0.3625.	K-Means had the highest silhouette score among the three tested methods.

Discussion. The EDA results show that demand changes by hour, season and weather. The association rules also connect season, evening time and clear weather with high demand. Classification results show that KNN gives the best overall score among the tested models. Clustering adds another view by grouping the 731 daily profiles into four K-Means segments with different average demand levels.

Comparative analysis. KNN achieved the highest classification accuracy (0.7178), while the Rule-Based model had the lowest (0.4876). Apriori and FP-Growth each returned 126 frequent itemsets and 89 rules; their recorded execution times are 0.010125 s and 0.056819 s respectively. Among the clustering methods, K-Means had the highest silhouette score (0.3625), followed by Hierarchical (0.3350) and DBSCAN (0.3235). DBSCAN also identified 66 noise days. The Davies–Bouldin values are included in Table 13.

11. Limitations, Societal Implications and Possible Future Enhancements

11.1 Limitations

- The data covers only two years and a single city, so findings may not generalise to other cities or climates.
- Records are aggregate counts per hour; individual trips, stations and routes are not available, so station-level imbalance cannot be studied.
- Weather attributes are normalized by the dataset authors, so original values must be recovered by the scaling factors.
- The demand classes depend on the chosen thresholds, and clustering results depend on the chosen features and parameters.
- The experiment is based on a two-year dataset from one city. Some execution details, including classifier training times, DBSCAN MinPts and PCA explained variance, were not present in the supplied result files and are therefore not reported.

11.2 Societal implications

- Accurate demand forecasting improves bike availability, which encourages low-carbon transport and reduces traffic congestion.
- Insights about weather and time-of-day demand help cities plan cycling infrastructure and safety measures.
- Fairness should be considered: models trained on historical usage may under-serve areas with historically low usage, reinforcing inequality in access.
- Although this dataset contains no personal information, trip-level data in real systems can reveal individual movements, so privacy protection is essential.

11.3 Possible future enhancements

- Regression and time-series models (for example gradient boosting or LSTM) to predict the exact rental count.
- Inclusion of station-level data, events, and pricing for finer redistribution planning.
- Ensemble classifiers and hyper-parameter optimisation.
- A real-time dashboard connected to the warehouse for operators.

12. Conclusion and References

12.1 Conclusion

This project analysed the UCI Bike Sharing Dataset through exploratory analysis, preprocessing, a SQLite data warehouse, classification, association mining and clustering. The cleaned dataset contained 17,379 hourly records with no missing values or duplicate rows. The OLAP outputs provided seasonal and time-based rental summaries. KNN achieved the highest classification accuracy of 0.7178 and F1-score of 0.7104. Apriori and FP-Growth each generated 126 frequent itemsets and 89 rules. K-Means grouped 731 daily profiles into four clusters and achieved the highest silhouette score (0.3625) among the tested clustering methods. The results show that rental demand varies with time, season and weather, and that these patterns can support planning for bike availability.

12.2 References

1. H. Fanaee-T and J. Gama, "Event labeling combining ensemble detectors and background knowledge," Progress in Artificial Intelligence, vol. 2, no. 2–3, pp. 113–127, 2014.
2. H. Fanaee-T, "Bike Sharing Dataset," UCI Machine Learning Repository, 2013. https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset
3. J. Han, M. Kamber and J. Pei, Data Mining: Concepts and Techniques, 3rd ed., Morgan Kaufmann, 2012.
4. R. Agrawal and R. Srikant, "Fast algorithms for mining association rules," in Proc. 20th VLDB Conference, 1994, pp. 487–499.
5. J. Han, J. Pei and Y. Yin, "Mining frequent patterns without candidate generation," in Proc. ACM SIGMOD, 2000, pp. 1–12.
6. M. Ester, H.-P. Kriegel, J. Sander and X. Xu, "A density-based algorithm for discovering clusters in large spatial databases with noise," in Proc. KDD, 1996, pp. 226–231.
7. F. Pedregosa et al., "Scikit-learn: Machine learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.
8. S. Raschka, "MLxtend: Providing machine learning and data science utilities and extensions to Python's scientific computing stack," Journal of Open Source Software, vol. 3, no. 24, p. 638, 2018.
9.

13. Appendix – Supplementary Results and Confusion Matrices

The following confusion matrices are reproduced from the model evaluation output. Rows show actual demand classes and columns show predicted demand classes.

Confusion Matrix – Decision Tree

Actual / Predicted	Low	Medium	High
Low	721	223	27
Medium	48	782	64
High	1	991	619

Confusion Matrix – Rule-Based

Actual / Predicted	Low	Medium	High
Low	272	632	67
Medium	0	800	94
High	0	988	623

Confusion Matrix – KNN

Actual / Predicted	Low	Medium	High
Low	796	118	57
Medium	182	584	128
High	81	415	1115

14. Presentation and Demonstration Supporting Materials

Table 16: Suggested demonstration outline

Step	Content	Time
1	Problem statement and dataset	1 min
2	EDA findings and preprocessing	2 min
3	Warehouse schema and OLAP demonstration	2 min
4	Classification comparison	2 min
5	Association rules and clustering	2 min
6	Conclusion and future work	1 min